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Are you in the room? A framework for sortition and representation

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Table of contents

Copyright notice iv

Abstract v

Declaration vi

Acknowledgements ix

1 Introduction 1

2 Literature Review 4
- 2.1 Representation and Sortition . . . 4
- 2.2 Panel Selection . . . 5
- 2.3 Visualisation Methods . . . 10

3 Methodology 12
- 3.1 Research Aims . . . 12
- 3.2 Research Design . . . 12

4 Results 18
- 4.1 2-variable visualisations . . . 18
- 4.2 Panel construction . . . 21
- 4.3 Framework and recommendations . . . 21

Bibliography 25

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Abstract

Sortition, the random selection of individuals to form panels, has seen a sort of renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

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I would like to thank my pet goldfish for ...

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Chapter 1

Introduction

In a social landscape where the public is growing progressively disillusioned with its representation in politics (Landemore 2026), it is important to consider ways in which we can foster more inclusive decision-making practices. One way of achieving this is using sortition to create panels that are more representative of the population for political discussions and deliberations. Adopted recently by an increasing number of institutions, including the Parisian ("Paris Creates a Permanent Citizens' Council" 2021), Irish (Walsh & Elkink 2021), and British Columbian (Ward 2008) governments, sortition as a part of democratic process is gaining traction. As a result, the need for fair and representative methods of building panels is only growing, and progress is being made in improving the structure of selective algorithms. However, there are still many challenges to overcome in order to ensure that these panels consistently achieve the level of representation that they envision. And there remains little procedure on measuring representativeness beyond the "surface" marginal level of demographics.

Sortition is a method of selection that uses random sampling to create panels that should be descriptively representative of a population. It has been used in various contexts, reaching as far back as ancient Athens (where it was how jurors were selected), and now into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative and inclusive than those selected through traditional methods, like elections or open-ended surveys. Due to its random nature, sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), making it more likely that decisions are in the relevant population's best interest.

One of the main roadblocks in sortition is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of what it means to be fair, what it means to be representative, and how

these look from a systematic perspective.

As a process, sortition can be broken down into four stages: 1) defining the population and traits of interest, 2) requesting expressions of interest from a random subset of the population, 3) feeding respondents into a sortition algorithm, and 4) panel selection and engagement for deliberation. Notably, there is currently no rigorous evaluation of representation achieved before a process is run. Additionally, each of these steps has the potential to introduce bias, impacting the representativeness of the panel. As such, it is crucial to measure how the level of representativeness (or *potential* representativeness) flows through each stage, and to understand how different factors may affect it.

Sortition methods have been evolving over time. Prior to algorithmic means, panel selection was carried out through simple self-selection (response to survey/advertising) and proportional sampling to meet panel size requirements. As sortition began to grow more robust, the Legacy techniques (Flanigan et al. 2021) started surfacing—these saw panels put together through simple choosing of respondents who met the most required "category" condition at any given moment, and would continue iterating until the required marginal counts were present in the selected panel. This led to situations of extreme selection likelihoods, if minorities existed—as members were selected indiscriminately of implied likelihood, meaning there were situations where an individual would potentially be selected 0% of the time. To counteract this, the more systematic algorithm "MaxiMin" was implemented (Baharav & Flanigan 2024); seeking not only to meet panel quotas, but to raise the lowest individual's selection likelihood, with the aim of eliminating the 0% selection case, in order to improve representativeness. This, however, now presented the risk of likelihood manipulation, as selection was based more directly on the combination of all category responses given by an individual, rather than pure counts needed for the panel to be filled. As a result, the inverse function "MiniMax" was employed to reduce extreme cases on the other end—eliminating the 100% selection chance case by minimising maximal selection likelihood of any given individual, providing an approach that was more resistant and robust to coercion, but simultaneously reintroducing the potential 0% likelihood problem. Though robustness itself lies beyond the scope of this paper's aims, it is important to note the existence of MiniMax as an approach to the sortition problem, as it impacted further algorithmic development.

More recently, and of direct relevance to the current study, further iterations of MiniMax and MaxiMin, in the forms of "LexiMin" (Flanigan et al. 2021) and "Goldilocks" (Baharav & Flanigan 2024), were established. These present more generally practical outcomes, with the former being used in industry at present. LexiMin takes the idea of MaxiMin and repeats it for every individual sequentially, maximising *all* individuals' likelihoods, rather than just those who sit at the absolute bottom. Thus, raising the

selection likelihood floor for the participant pool overall. It is, however, like MaxiMin, vulnerable to manipulation (which remains beyond the scope of this paper)—leading into the development of the Goldilocks algorithm solution. Goldilocks aims to simultaneously raise the selection likelihood floor (eliminating the 0% selection chance and improving representativeness), as well as lower the ceiling (eliminating the 100% selection case and making the process more robust against manipulation). A tool for Goldilocks' use is in development at the time of the writing of this paper, and is expected to be made publicly available in the near future. As such, it will not be used for the current study, but may present an opportunity for further research.

Notably, all of these algorithms only focus on optimising for maximal selection likelihoods as their sole objective functions. This means that in its discovery and construction of feasible panels, the existing LexiMin algorithm may be prone to distortion of intersections and intersectional association trends present within population distributions. This presents a massive risk for the validity of representation achieved by panels output by the algorithm–the extent and reasoning of which will be explored further in the paper.

Additionally, there is limited literature on the visualisations of the outputs of these algorithms. This presents a great opportunity to demistify LexiMin and sortition as a process, improving transparency for practitioners, facilitators, participants, and broader community stakeholders alike. The comparative visualisation methods and tools also allow a general comparison to be made between any selected group and any broader sample it is pulled from, with any algorithm or selection process. This could be massively useful as a mechanism for quick sanity checks of panel configurations and as a way for practitioners to evaluate the validity of panels before engaging in deliberation.

The remainder of this thesis is structured as follows: Chapter 2 provides a literature review of the existing research on sortition and representation, and identifies gaps in the current knowledge. Chapter 3 outlines the methodology used to evaluate the representativeness of panels and to investigate the impact of unobserved traits and survey non-response. Chapter 4 presents the results of the analysis, and Chapter 5 discusses the implications of the findings and suggests directions for future research.

Chapter 2

Literature Review

2.1 Representation and Sortition

2.1.1 What does it mean to be representative?

The main aim of any and all selection methods within sortition is to create a panel that is fair and representative of the population. But what does this actually mean?

Fairness refers to the idea that the selection process should be transparent, unbiased, and inclusive. It should give everyone an equal chance of being selected, regardless of their background or characteristic, which presents a likelihood difference minimisation challenge. Systematically, this is typically achieved through introducing randomness to selection processes (Flanigan et al. 2021), as not doing so immediately introduces implicit selection bias.

Alongside this, representativeness refers to the idea that the selected panel should accurately reflect the demographics and perspectives of the population from which it was built. This means that the panel should be diverse and inclusive, and should contain individuals from a range of prediagnosed backgrounds of interest. This can be assessed directly by comparing demographic compositions of the selected panels against the population from which it was originally drawn.

Approaching it systematically, the sortition process can be thought of in four stages:

- Defining the population and traits of interest
- Sampling from the population (through survey or other outreach)
- Panel selection (through algorithmic means or other method)
- Panel evaluation, confirmation and engagement

Each of these steps presents an opportunity for bias to be introduced, and for the panel to become less representative of the population. Which is why it is crucial to evaluate the level of representativeness (or *potential* representativeness) at each stage, and to understand how different factors may affect it.

For example, if the population of interest and the investigated traits are poorly defined, then the selection process may not be able to capture the diversity and perspectives of the desired group, leading to a less representative panel. This presents a challenge in itself, given most population descriptions (as previously discussed) are often based on disjoint demographic characteristics and do not provide a way to capture intersectionality, which is a key aspect of representativeness. However, this may be beyond the scope of this thesis, as it is already a foray into broader data collection and storage methods, but it is important to keep in mind that representation goes beyond the margin.

Similarly, if certain groups are less likely to respond to the survey, then they may be underrepresented in the pool of potential panelists, which can lead to a less representative panel from the outset. Illustratively–if there were to be no female respondents for a certain survey in an area comprised of 60% women, then the representativeness assumption (and goal) would be majorly violated before an attempt at putting a panel together has even been made. Situations like these raise the question of how we can account for this non-response bias, and whether we can use visualisation tools to show the discrepancy between the pool and the population (which leads back to part of this thesis' aim).

2.2 Panel Selection

2.2.1 Problem Definition

Suppose there is a pool of individuals numbered 1 to n.

We want to choose a panel of size k.

Let K be the set of all feasible panels. A feasible panel is any group of k individuals that satisfies the required constraints (which generally take the form of quotas).

These quotas will be certain desired counts, or ranges of counts, of individuals belonging to a certain trait class (e.g. 20-22 males, or 15-17 rural). These traits and quotas, by extension, will have direct ties to the demographics that have been deemed crucial to represent for a certain project's population group. Tying back to the established background–the core goal of sortition, then, is to build a panel which meets the quota constraints, equitably, robustly, and in a way that represents the population from which it was drawn.

2.2.2 Existing Methods

Non-algorithmic/self-selection

This is the most basic method of panel selection, where individuals are simply allowed to self-select into the panel. This can be done through a variety of means, such as advertising the opportunity to participate in the panel and allowing individuals to sign up on a first-come, first-served basis. However, these types of methods are often not very effective at creating representative panels, as they can a) lead to self-selection bias (as involvement will be tied to exposure and not random chance, those who put their hand up of their own volition are more likely to have pre-formed perspectives, ahead of the panel process), thus possibly not capturing the diversity and perspectives of the population, and b) lead to lower-than-desired engagement, as it does not guarantee any particular level of exposure or direct engagement in the first place.

LEGACY heuristic (I can't find original source, but it is mentioned in Flanigan et al. (2021))

Already a systematic improvement over non-algorithmic methods; the LEGACY selection heuristic takes a given sample of respondents and selects individuals one-by-one until the panel is full. At each step, the heuristic calculates a "need" value to identify the trait that is currently furthest from the desired count in the panel. LEGACY then selects a random individual from the pool who possesses that trait. If selecting any given individual would cause the panel to hit the upper quota for any trait, then all other individuals with that trait are removed from the pool before the next selection. This process is repeated until it either becomes infeasible (exhausts the dataset before hitting overall quota, at which point it will restart), or builds a full panel with desired quota representation. LEGACY's process fulfils the base requirements of building a panel that meets the desired quotas, but it does not necessarily maximise representativeness, as it does not take into account the likelihood of selection for different individuals (allowing them to just be 0%, in extreme cases), and it does not attempt to minimise the maximum deviation from equal selection chance across all individuals.

MaxiMin and MiniMax (Baharav & Flanigan 2024)

This aforementioned shortcoming is then addressed by our first algorithmic sorition methods: MaxiMin and MiniMax, though these both tackle the problem in opposite ways and with differing goals. MaxiMin, as the name would suggest, is a selection algorithm that generates potential panels and selection probabilities and then sets out to maximise the minimum selection probability of the set. This reduces the likelihood of an individual never being chosen (improves fairness) but makes the method quite susceptible to manipulation, if an individual elects as belonging to certain quota groups with lesser representation in the sample. Thus, leading to potential representative burden. Inversely, MiniMax aims to minimise the maximum probability of an individual being selected, with the aim of making selection more robust (less vulnerable to manipulation). However, this then comes at the

cost of fairness, as the issue of potential 0% selection likelihood returns.

LexiMin (Flanigan et al. 2021)

LexiMin, or lexicographic minimisation, is a method of panel selection similar to MaxiMin, that aims to maximise the minimum likelihood of selection, but for each individual being selected sequentially (rather than just the outright minimum). It works by first identifying the individual with the lowest selection chance, and then selecting panel combinations that make this individual's selection chance as high as possible. This process is repeated for each individual in order of their selection likelihood until overall selection chance is maximised and, ideally, made as equal as possible across all participants. This ensures that each individual has the highest possible chance of being selected. But this trait of LexiMin can, like MaxiMin, make it vulnerable to manipulation.

Goldilocks (Baharav & Flanigan 2024)

Bringing us to the supposed current gold standard: the aptly named "Goldilocks" sortition algorithm. Goldilocks is structured in such a way that it simultaneously attempts to maximise minimum selection probabilities (improve fairness) and minimise maximum selection probabilities (improve robustness), providing a "best of both worlds" solution to the problem by averaging outcomes.

2.2.3 LexiMin Approach (Flanigan et al. 2021)

LexiMin does not choose one panel immediately. Instead, it chooses a probability distribution over all feasible panels.

Each feasible panel is assigned a probability.

These probabilities must be non-negative and must add up to 1.

For each individual, we then calculate their overall probability of being selected by adding up the probabilities of all feasible panels that include that individual.

Next, we sort the individual selection probabilities from smallest to largest.

The LexiMin rule works by first making the smallest selection probability as large as possible. Subject to that, it makes the second-smallest as large as possible. It then does the same for the third-smallest, and continues in this way.

2.2.4 Formal Representation

Let p_S be the probability assigned to panel S.

These probabilities satisfy:

$$p_S \geq 0$$

and

$$\sum_{S \in K} p_S = 1$$

The selection probability for individual i is:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

This gives the vector of individual selection probabilities:

$$\pi(p) = (\pi_1(p), \pi_2(p), ..., \pi_n(p))$$

Sort these from smallest to largest:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq ... \leq \pi_{(n)}(p)$$

LexiMin chooses p so that this sorted vector is maximised in order.

2.2.5 Algorithm

1. List all feasible panels in K.
2. Consider all possible probability distributions over those panels.
3. For each distribution, compute:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

4. Sort the selection probabilities:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq ... \leq \pi_{(n)}(p)$$

5. Choose the distribution that maximises the smallest selection probability.

6. If more than one distribution still remains, choose the one that maximises the second-smallest selection probability.
7. Continue in this way until the distribution is determined.
8. Sample a final panel according to that distribution.

2.2.6 Logic

LexiMin is a fairness rule. It gives priority to the individuals with the lowest chance of being selected and improves their chances first. Then it improves the next-lowest probabilities, and so on.

This makes the selection probabilities as equal as possible while still respecting the panel constraints.

2.2.7 Discussion

This approach to the problem of panel selection is elegant and has some nice properties. It ensures that the selection probabilities are as equal as possible, which can help to create a more representative panel. However, it does not give a way to introduce specific joint trait requirements for subsets of the population, which is something that may be desirable in certain situations. For example, if there is a need to ensure that a certain combination of traits is represented in the panel, then this approach may not be able to accommodate that requirement without reintroducing representative burden.

2.2.8 New Investigation

Across all algorithmic techniques, it is noted that stratification for combinations of demographic traits is not included within their decision-making designs–focusing instead purely on affecting individual selection chances based on quotas of each demographic trait at the margin. This is a major limitation, as it means that the selection process may not be able to entirely capture the diversity and perspectives of the desired group, leading to a less representative panel. In some situations (e.g. supporting side-panels built for specific subgroups, communities where there are known intersectional minorities of interest), it may be beneficial, or even necessary, to ensure the inclusion of specific intersectional combinations of traits in order to truly represent a given population. Though this limitation is generally mitigated in practice by bringing in “specialist speakers” to talk to the panel during their deliberation processes (Redman & Carson n.d.), this still is not the same type of inclusion by nature, as being directly part of the panel and decision-making process.

The extent to which intersectional inclusion happens naturally through current methods will be evaluated in the later simulations. We will then also test the effects of further imposing combinational quotas on the representative burden, and on the selection probabilities of different individuals, to see

whether this is a feasible way to ensure the representation of intersectional minorities in situations where it is deemed necessary.

2.3 Visualisation Methods

In order to evaluate the representativeness of selected panels, we will use a variety of visualisation methods to show how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process. Some potentially useful visualisation methods include: bar charts, mosaic plots, double decker plots, UpSet plots and parallel coordinate plots.

First, it will be of use to just do some illustrative graphing to show densities within the distribution across traits, in the form of bar charts. These will give a rough visual idea of the breakdown of the population and samples within the simulation, setting up some expectations for how representation should look in further visualisations. From there, either mosaic or double decker plots will be useful for depicting the proportion of given demographic traits that are selected for the panels. Finally, UpSet plots (Conway, Lex & Gehlenborg 2017) will be an effective way to showcase the intersections that exist in generated panels, as they offer a visual link between demographic variables, as well as frequency counts to highlight how often these occur within a given panel.

– ignore the summaries –

2.3.1 Parallel coordinate plots (VanderPlas et al. 2023)

- Can help to show the distribution of traits across the selected panel and how it compares to the overall population.
- Can also show the trade-offs between different traits and how they are balanced in the selection process
- Can be used to identify any segments of the population that are underrepresented in the selected panel, and to highlight any potential biases in the selection process.

2.3.2 Matrix-based set representation (UpSetR) (Conway, Lex & Gehlenborg 2017)

- Can be used to show how different traits connect with each other and how they are distributed across the distribution and/or selected panel.
- Can help to identify any clusters of traits that are overrepresented or underrepresented in the selected panel, and to show how different traits interact with each other in the selection process.

- Can highlight specific combinations of traits that are of interest, and show how they are represented in the selected panel compared to the overall population. (Though there is no way to gauge the prevalence of these combinations in the population, beyond anecdotal observations)

–

2.3.3 Attribute distribution analysis

- Showcase how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process.

Chapter 3

Methodology

3.1 Research Aims

What does it mean for a panel to be “good” at being representative and are we achieving this? - Are current methods and algorithms, like LexiMin, performing the best they could be? - How do unobserved/unfiltered traits affect response rates, and therefore panel composition and representativeness? - How can survey non-response and other sampling-level issues bias representativeness?

nDF suggests 10x response to panel count; not generally feasibly, more often get 3x. How much of a difference does this make to burden and selection likelihood?

3.2 Research Design

3.2.1 Simulation design

This thesis investigates the impacts of data complexity and construction on algorithmic performance. To test this, populations of varying intersectional complexity (three and five categories) and varying associations (independent and associated) were simulated. Categories used were inspired by demographic variables often used for sortition in industry. In the three categorical variable case, these were age (taking values of 18-34, 35-44, 45-54, 55-64, 65+), gender (taking values of male, female, non-binary) and tenure (taking values of rent, own, other). The five variable case took the initial three variables from the previous case, but then also added the variables education (no qualification, secondary, bachelor, postgraduate) and location (metro, regional, remote). The combinations of these variables were then used to produce the simulated datasets for the panel sampling process.

To illustrate the general procedure concretely at a minimal scale, a simplified two-variable example was constructed using only age and tenure. Age is denoted A, taking one of $L_A = 5$ ordered levels $a \in \{1, \ldots, 5\}$ (18-34 through 65+), as previously stated, and tenure is denoted T, taking one of $L_T = 3$ levels $t \in \{1, \ldots, 3\}$ (Rent, Own, Other). Marginal probability vectors were first defined for each variable,

$$\mathbf{p}_A = \left(p_{A_1}, \ldots, p_{A_5}\right), \qquad \mathbf{p}_T = \left(p_{T_1}, p_{T_2}, p_{T_3}\right), \qquad \sum_{a=1}^{5} p_{A_a} = 1, \qquad \sum_{t=1}^{3} p_{T_t} = 1,$$

with $\mathbf{p}_A = (0.15, 0.20, 0.25, 0.25, 0.15)$ specified as a mild skew toward the upper-middle age brackets, and $\mathbf{p}_T = (0.30, 0.50, 0.20)$ specified with a moderate skew toward home ownership. The same marginal targets $\mathbf{p}_A$ and $\mathbf{p}_T$ are used for both the independent and associated populations described below, so that any difference between them is attributable solely to the association introduced, rather than to a difference in either variable's marginal distribution.

An independent population (`sim_data2`) was generated by taking the outer product of these two marginals to form a joint probability array,

$$P^{(0)}(a, t) = p_{A_a} p_{T_t}, \qquad \mathbf{P}^{(0)} = \mathbf{p}_A \mathbf{p}_T^{\top} \in \mathbb{R}^{5 \times 3},$$

a 5×3 matrix whose row sums equal $\mathbf{p}_A$ and column sums equal $\mathbf{p}_T$ by construction, so that $A \perp T$. This array was flattened into a probability vector and cell counts were drawn from a single multinomial trial before being expanded into individual-level records, as described in full below.

A second, associated population (`sim_data2_cov`) was then generated using the same marginal targets, but with a negative association deliberately introduced between age and renting. This required an ordinal score for age,

$$s(a) = \frac{a - 1}{5 - 1} \in [0, 1],$$

running from 0 for the youngest bracket to 1 for the oldest, which was used to seed the independent array $\mathbf{P}^{(0)}$ with an exponential adjustment factor targeting only the "Rent" level of tenure,

$$\tilde{P}(a, t) = P^{(0)}(a, t) \cdot \exp\Big(-\alpha\, s(a) \cdot \mathbb{1}[t = \text{Rent}] \Big), \qquad \alpha = 1.2,$$

so that the probability of renting is scaled down by a factor of $e^{-\alpha s(a)}$ that grows with age, while the probability of owning or other tenure is left unchanged. Because this adjustment is applied only to the "Rent" column, the row and column sums of $\tilde{\mathbf{P}}$ no longer equal $\mathbf{p}_A$ and $\mathbf{p}_T$, so the array was subsequently rescaled through iterative proportional fitting (raking) so that the age and tenure marginals were restored exactly to their specified targets. Initialising $P^{(0)}_{\text{rake}} = \tilde{P}$, each raking iteration $r = 0, 1, 2, \ldots$ performs two rescaling passes,

$$P^{(r+1/2)}(a,t) \;=\; P^{(r)}_{\text{rake}}(a,t) \cdot \frac{p_{A_a}}{\sum_{t'} P^{(r)}_{\text{rake}}(a,t')}, \qquad P^{(r+1)}_{\text{rake}}(a,t) \;=\; P^{(r+1/2)}(a,t) \cdot \frac{p_{T_t}}{\sum_{a'} P^{(r+1/2)}(a',t)},$$

first rescaling every row to sum to p_{A_a}, then every column to sum to p_{T_t}. Repeating this pair of passes for $R = 100$ iterations (well past convergence for a 5×3 table) yields the raked array $P^\star$, satisfying

$$\sum_t P^\star(a,t) = p_{A_a} \;\text{ for all } a, \qquad \sum_a P^\star(a,t) = p_{T_t} \;\text{ for all } t,$$

to numerical tolerance, while retaining the negative age-renting association introduced by the exponential seeding step – this is the sense in which raking restores the specified marginals while preserving the association structure introduced by the adjustment.

Both populations were sampled the same way, differing only in which array was used. The relevant array ($\mathbf{P}^{(0)}$ for the independent case, or the raked $P^\star$ for the associated case) was flattened into a probability vector $\mathbf{p}^\star \in \mathbb{R}^{15}$ (age varying fastest), and cell counts were drawn from a single multinomial trial,

$$(n_1, \ldots, n_{15}) \sim \text{Multinomial}(n_{\text{total}}, \mathbf{p}^\star), \qquad n_{\text{total}} = 100{,}000,$$

with each cell j, corresponding to one (a,t) combination, expanded into n_j individual-level records sharing that age and tenure. Comparing the two resulting populations demonstrates the core logic of the simulation framework in its simplest form: a chi-squared test of association between age and tenure is non-significant in the independent population but highly significant in the associated population, with the conditional probability of renting falling steadily from around 44% among 18-34 year olds to around 18% among those aged 65 and over, even though both populations share identical marginal distributions for age and tenure individually. This confirms that the seeding-and-raking procedure successfully introduces a targeted association between variables without disturbing their

overall marginal proportions. This will allow for testing of the algorithm's ability to preserve sample intersection structures, beyond a purely marginal level.

Both populations also include an additional binary variable, generated independently of age and tenure for each simulated individual i via

$$Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5),$$

drawn after the population has been expanded to individual level, so that Z_i is uncorrelated with age and tenure by construction. This variable is deliberately excluded from all visualisations, representing an unobserved (latent) trait whose presence is known to the simulation but not exposed to any downstream algorithm. This was used to test whether unobserved traits can bias panel composition even when they play no role in the population's observed demographic structure.

Next, as is standard in practice, a random slice of defined proportions (in this case 10%) is taken from the population, to be used as the survey sample, from which responses are drawn. Concretely, from a population of n_{total} individuals, a survey sample $\mathscr{S}$ of size $n_{\mathscr{S}} = \lfloor 0.10\, n_{\text{total}} \rceil$ (so $n_{\mathscr{S}} = 10{,}000$ for the two-variable example, drawn uniformly at random without replacement) is taken as the pool of individuals invited to respond, before any response behaviour is simulated. Whether each invited individual $i \in \mathscr{S}$ actually responds is then modelled as an independent Bernoulli trial,

$$R_i \overset{\text{ind.}}{\sim} \text{Bernoulli}(\pi_i),$$

where the response probability π_i is assigned according to one of three cases, mirroring the independent, observed-trait, and latent-trait cases described above for the population itself.

In the *independent* case, every invited individual is equally likely to respond regardless of their trait profile,

$$\pi_i = \pi_0, \qquad \pi_0 = 0.30,$$

providing a baseline against which the effects of differential response can be compared – a chi-squared test confirms no significant association between A and R under this case, as expected. In the *age-covaried* case, response likelihood increases with age, reusing the same ordinal score $s(a)$ defined above,

$$\pi_i = \pi_1 \exp\big(\alpha_R \, s(a_i)\big), \qquad \pi_1 = 0.20, \quad \alpha_R = 1.0,$$

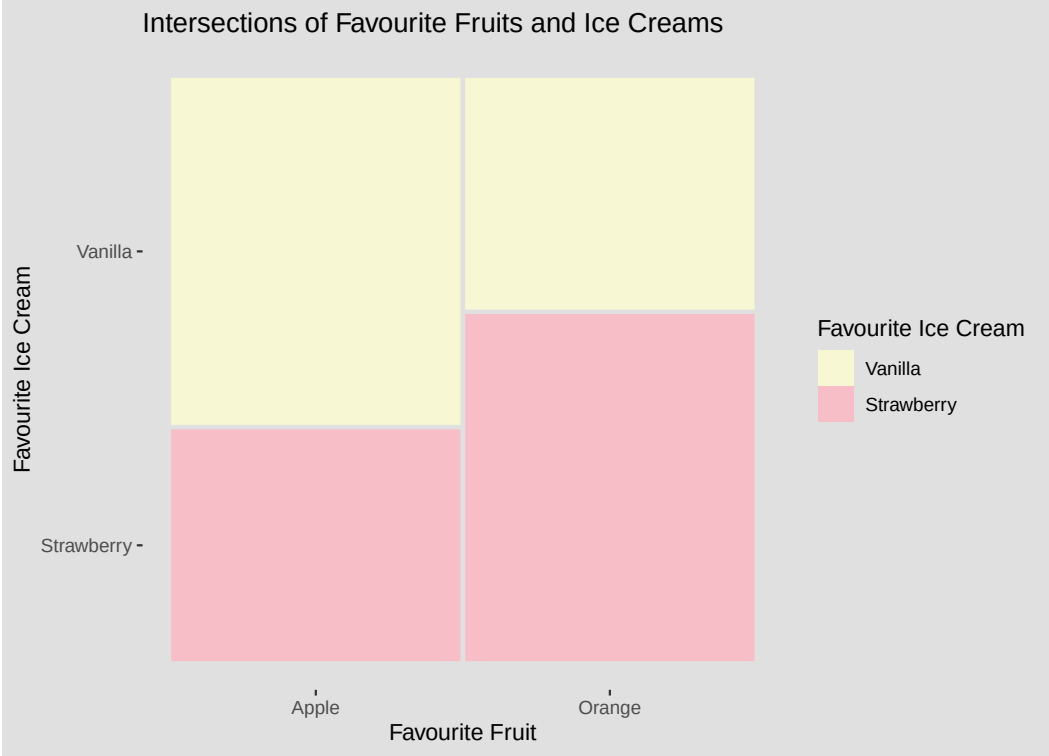
so that the response rate rises from around 20% among the youngest bracket to over 50% among the oldest, and the association between A and R is now highly significant. In the *latent-covaried* case, response likelihood instead increases with the unobserved trait Z_i rather than with any observed trait,

$$\pi_i = \pi_1 \exp\big(\alpha_Z \, Z_i\big), \qquad \pi_1 = 0.20, \quad \alpha_Z = 0.8,$$

giving a response rate of approximately 20% for $Z_i = 0$ and 44% for $Z_i = 1$. Critically, while the association between Z and R is highly significant under this case, the association between A (or T) and R remains non-significant, exactly as it was in the independent case. This is the central diagnostic point of the latent-covaried case: an unobserved trait can systematically bias who responds to a survey, and therefore who is available for panel selection, without ever manifesting as a detectable association in any of the demographic variables an algorithm can actually observe.

3.2.2 Visualisation methods

In order to read the later mosaic plots, a simpler plot can first be introduced.



For this plot, only two variables are examined: favourite fruit and favourite ice cream flavour. For the visualised example data set, 25 individuals with a preference for each marginal option were generated,

then shuffled randomly to make independent intersectional combinations. As such, the width of bars displays proportion of x-axis category responses conditional on y-axis responses (favourite fruit, conditional on favourite ice cream flavour), and vice versa. Fill colour is used to make the ice cream flavour preferences easier to read.

Chapter 4

Results

4.1 2-variable visualisations

4.1.1 Population simulation

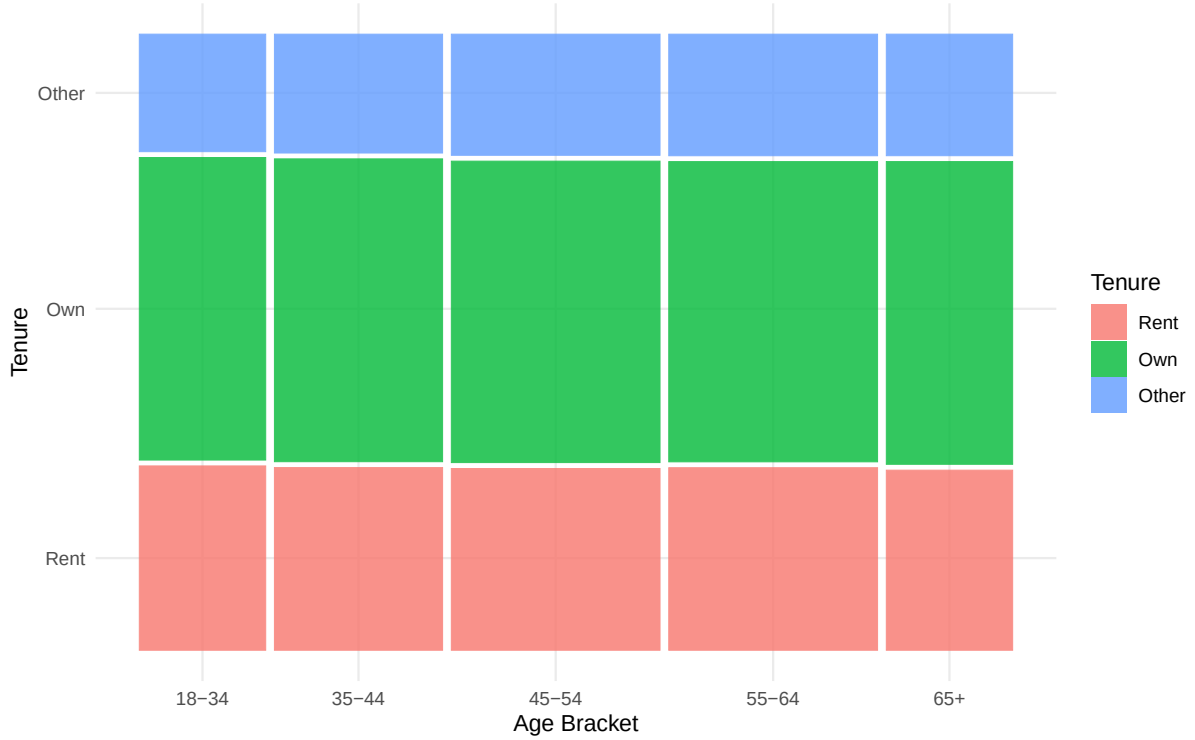


Figure 4.1: *Mosaic plot showing the intersection of age and tenure categories when they are generated independently, with age brackets on the x-axis and tenure categories on the y-axis and fill.*

Figure 4.1 and Figure 4.2 show the intersectional proportions of the age and tenure categories for the independently and associatively generated populations, respectively. Both skew towards upper middle age and higher frequency of homeowners (compared to other categories). But whilst there is

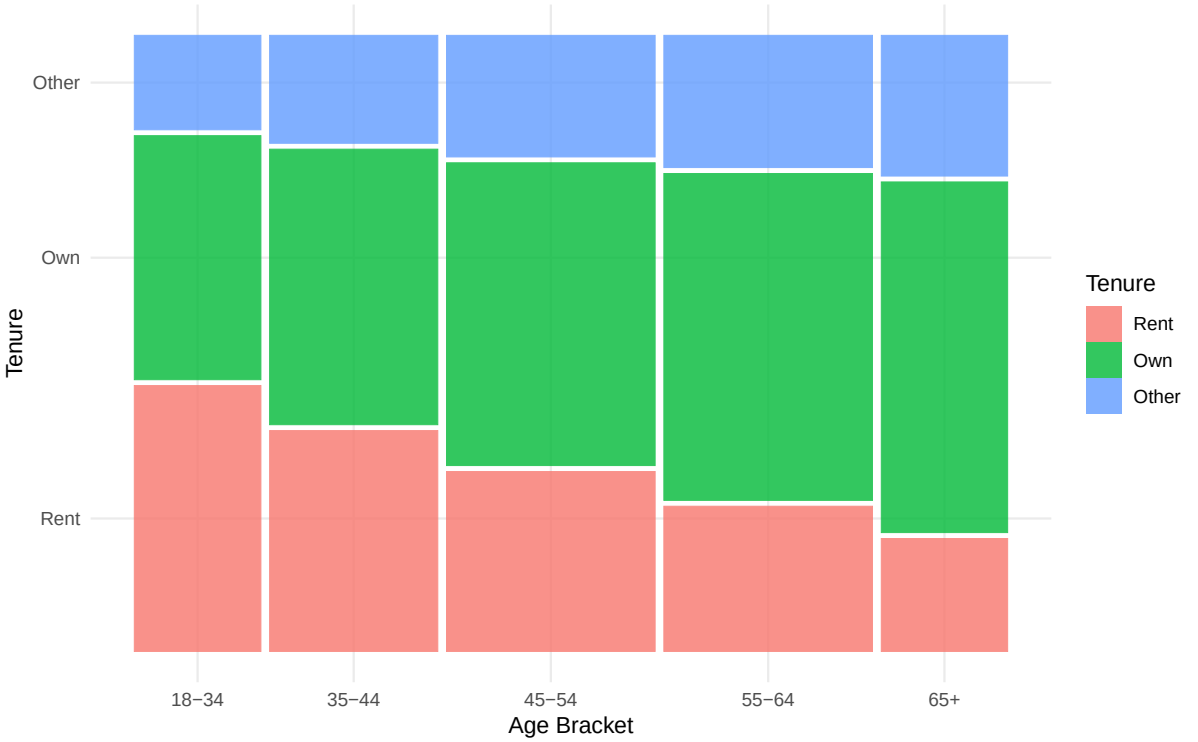


Figure 4.2: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting. Age brackets shown on the x-axis and tenure categories on the y-axis and fill.*

no visible association between categories in Figure 4.1, Figure 4.2 contains a negative association between age and renting. As such, the probability of any intersection in Figure 4.1 is explained purely by the margins, and Figure 4.2 probabilities vary by category.

4.1.2 Sample construction

Figure 4.3 and Figure 4.4 show random 10% (10000-individual) slices of the independent and associated populations respectively, replicating the mailout process of expressions of interest. Figure 4.3 shows Figure 4.1 's indepedendent structure is largely preserved through slicing, with very slight deviations in younger and older "Other Tenure" proportions. Figure 4.4 shows Figure 4.2 's associated structure is also preserved through slicing, with relative proportions maintained.

4.1.3 Respondent simulation

Figure 4.5 and Figure 4.6 show the intersections of the simulated respondents of the expression of interest survey, assuming an indepedently generated population. Figure 4.5 simulates responses with equal chance, while Figure 4.6 positive associations.

Figure 4.7

Figure 4.8

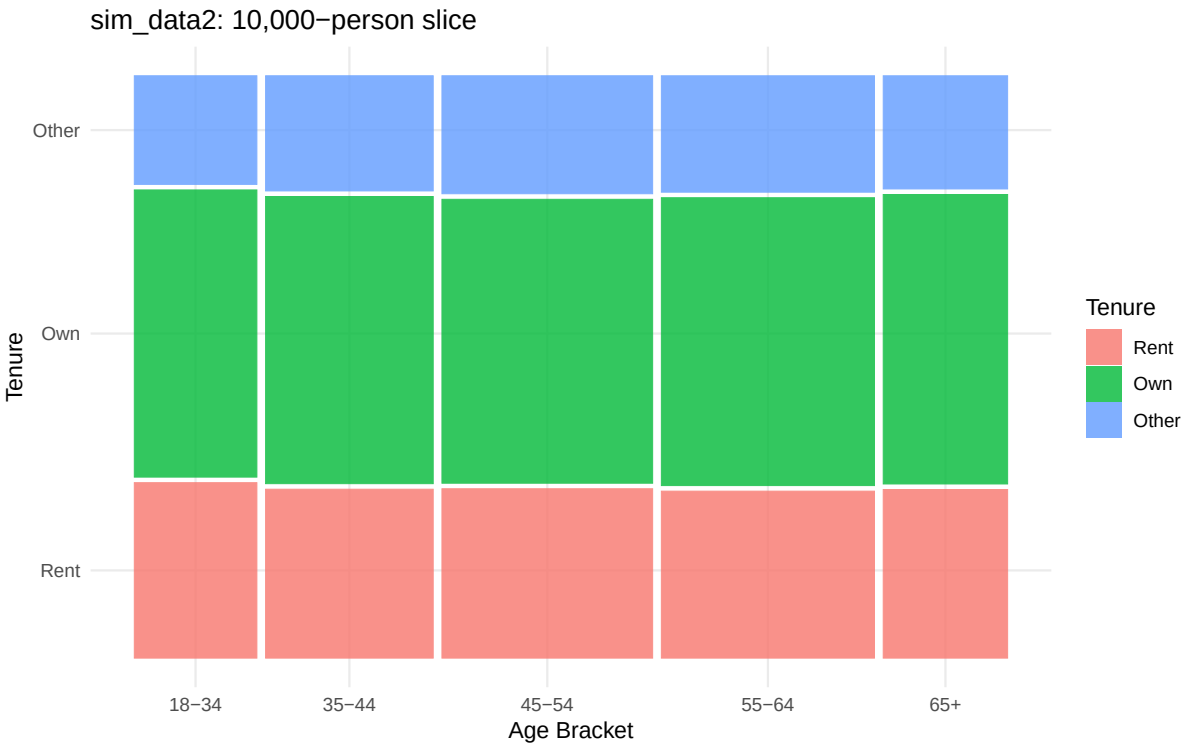


Figure 4.3: *Mosaic plot showing the intersection of age and tenure categories when they are generated independently for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

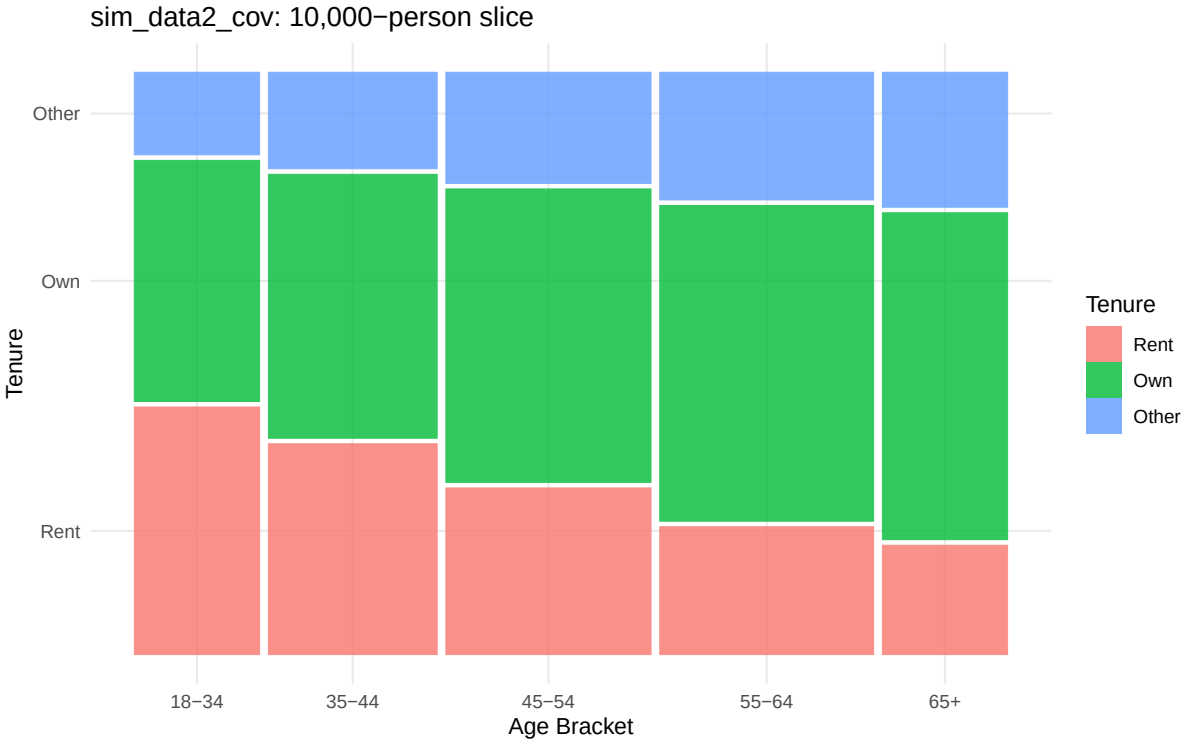


Figure 4.4: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

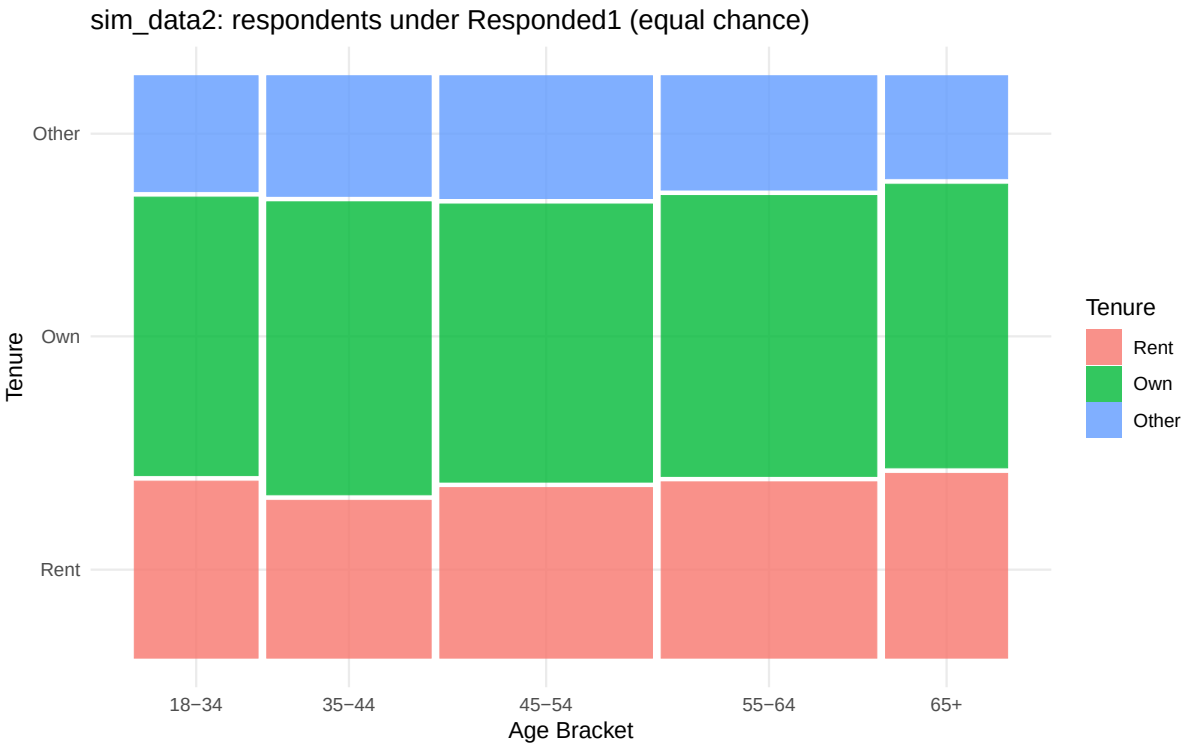


Figure 4.5: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

4.2 Panel construction

4.2.1 Association preservation

It is found that 100% of panels have a significantly different association level than exists

4.3 Framework and recommendations

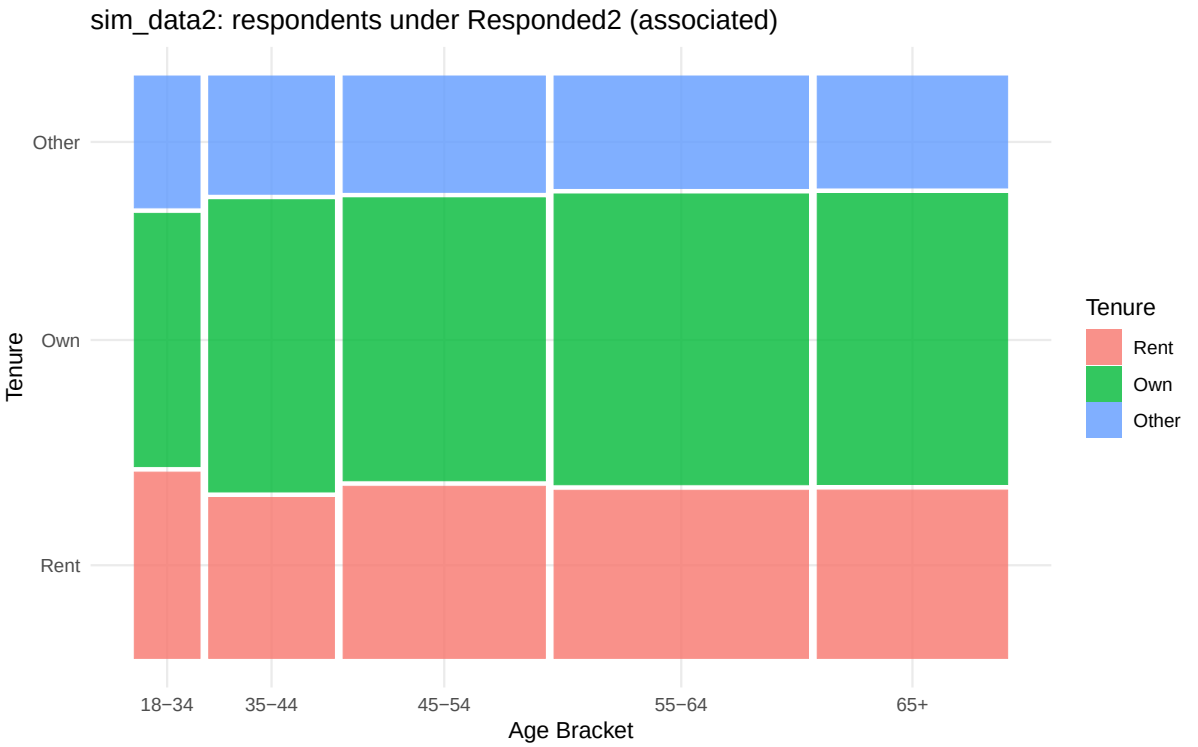


Figure 4.6: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

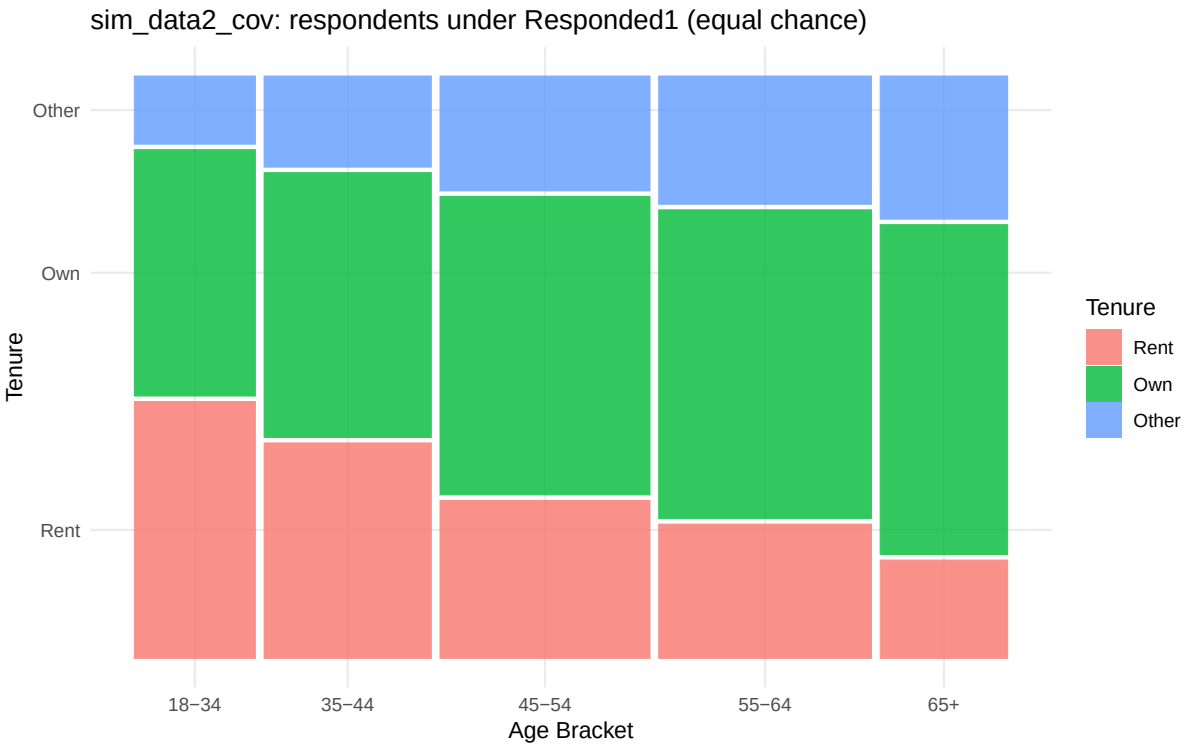


Figure 4.7: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

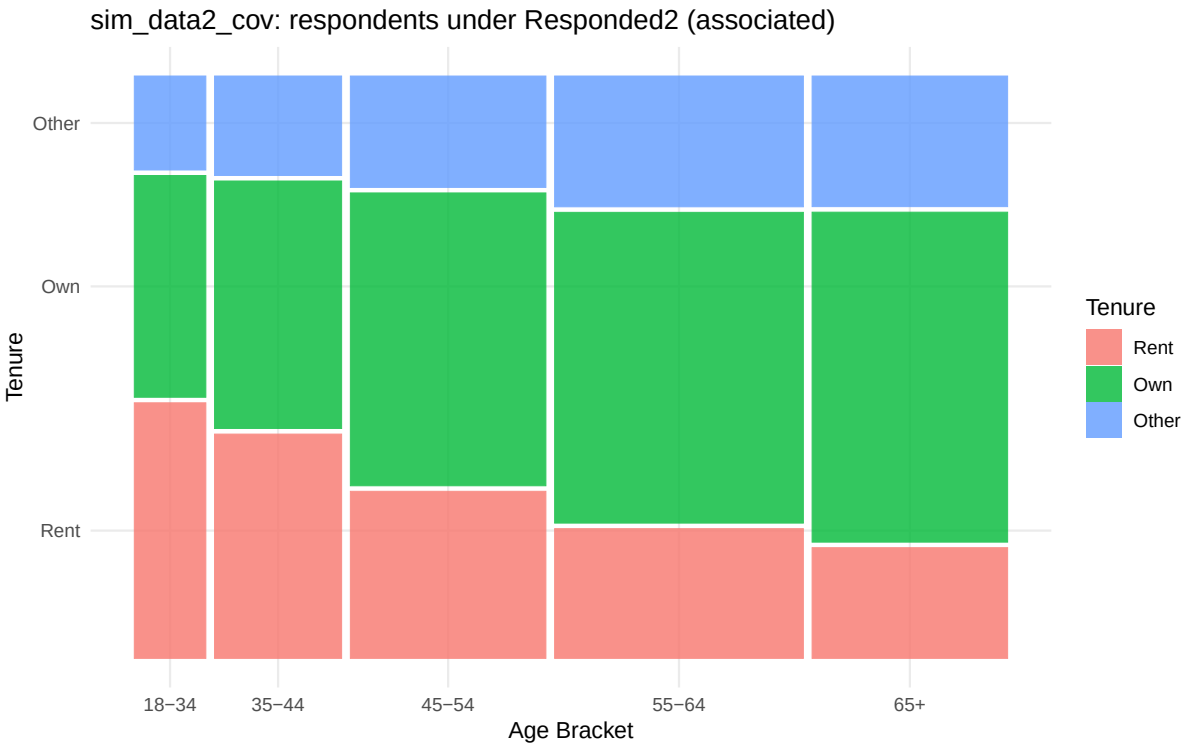


Figure 4.8: *Mosaic plot showing the intersection of age and tenure categories when they are generated with a negative association between age and renting for a 10000-individual slice. Age brackets shown on the x-axis and tenure categories on the y-axis and fill. Structure is preserved.*

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